Akshaj Gupta
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EDUCATION

University of California, Berkeley Expected Graduation: May 2028
B.S. in Electrical Engineering & Computer Science (EECS) Cumulative GPA: 4.0

RELATED COURSEWORK & SKILLS

Coursework: Artificial Intelligence, Robotics Systems, Circuits & Electronics, Data Structures & Algorithms, Computer Architecture & Machine Structures, Designing Devices and Systems, Discrete Math & Probability Theory, Linear Algebra

Languages: Java, Python, C++ **Skills:** Robotics, AI/ML/CV, Multimodal Learning

Activities: Stand-up Comedy Club **Tools:** PyTorch, TensorFlow, OpenCV, ROS, Git, Docker, YOLOv8, CAD

ACADEMIC HONORS AND AWARDS

UC Berkeley Regents' & Chancellor's Scholarship Recipient, 2024
National Merit Scholarship Recipient, 2024
5x American Invitational Mathematics Exam (AIME) Qualifier, 2023, 2022, 2021 Fall, 2021 Spring, 2020
Ranked Top 25 in the Nation for Public Speaking and Debate, 2024

WORK/RESEARCH EXPERIENCE

Researcher, Berkeley Artificial Intelligence Research (BAIR) Lab *January 2025 – Present*
Advised by Prof. Gopala Anumanchipalli in Berkeley Speech Group, Collaboration with Meta FAIR

- **Audiovisual Perception (with Meta FAIR):** Developing a multi-task audiovisual perception model using WavLM and Open-MAGVIT2 encoders. Implementing fusion pipelines and evaluating model performance on AV speech, emotion, speaker detection, intent, and other tasks across Candor, MultiDialog, MIntRec2.0, etc. datasets
- **Guitar Transcription:** Proposed and leading development of a four-stage ML pipeline converting guitar videos into sheet music through audio-to-MIDI pitch-based transcription models, expressive technique classification, transformer-based fret/string mapping, and automated tab generation across GuitarSet, EGDB, etc. datasets
- **LCS-CTC:** Helped develop a two-stage phoneme transcription framework combining similarity-aware alignment (via modified Longest Common Subsequence over frame-phoneme cost matrices) with constrained CTC decoding. Helped train alignment models, improving phoneme recognition on both fluent and disfluent speech (LibriSpeech, PPA, LLM-dys). **Co-authored** "LCS-CTC: Leveraging Soft Alignments to Enhance Phonetic Transcription Robustness"

Software Engineer Intern, Multiply Labs *June 2024 – August 2024*

- Contributed to development of robotic system that automates cell therapy manufacturing processes
- Designed and implemented a Python-based auto-calibration and data acquisition tool to accurately predict object positioning and calculate acceptable calibration error margins for safer robotic operations
- Assisted in R&D efforts by conducting research experiments assessing robotic versus manual production

Software Engineer Intern, Level 42 AI *June 2023 – October 2023*

- Developed an autonomous robotic arm for medical diagnostics capable of identifying wounds (e.g., lacerations, burns) and diseases (e.g., smallpox, skin cancer). Utilized custom YOLOv8-based computer vision models (91% wound detection accuracy, 95% anatomical classification accuracy) and designed the full data acquisition pipeline, including image capture, anatomical localization, wound detection, and 3D visualization on a human body model
- Presented project to **Bill Gates Global Health Lab, Berkeley California Life Sciences Innovation Showcase, and U.S. Air Force 59th Medical Wing**

Robotics Engineer Intern, NXP Semiconductors *May 2020 – November 2021*

- Built two autonomous robots that traverse the NXP office and run Wi-Fi tests for controlled testing of devices
- Designed and fabricated sensor mounts, robot chassis, and electronics enclosures using CAD

COMMUNITY SERVICE/LEADERSHIP

Advisor & Co-Founder, AlgoSTEM Inc. *March 2020 – Present*

- Co-founded a 501(c)3 non-profit organization that provides free STEM education to students worldwide
- Co-created **12** Udemy courses on teaching math, physics, & computer science with **100,000+** students. Co-created a YouTube channel with **2,200+ subscribers** and **122,500+ views** with solutions to problems from programming sites
- Received **$165,000+** from **Google**, **Meta**, **Jane Street**, **WolframAlpha**, **Art of Problem Solving (AOPS), Chipotle, etc.**